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Introduction

Why every business is now a data business

Data is changing our world and the way we live and work at an unprecedented rate. Depending on your viewpoint, we're either at the start of something incredibly exciting or we're entering a terrifying Big Brother era where our every move can be tracked – and even predicted. And both sides have a point. Business leaders and managers, however, have little time for data scepticism. Data is already revolutionizing the way companies operate and it will become increasingly critical to organizations in the coming years. Those companies that view data as a strategic asset are the ones that will survive and thrive. With the massive growth in big data and the Internet of Things, plus rapidly evolving methods for analysing data, the importance of data across every area of business will only grow.

The astonishing growth of data, artificial intelligence and the Internet of Things

Every two days we create as much data as we did from the beginning of time until 2003. Every two days! And the amount of data we're creating continues to increase rapidly. In fact, IDC predicts¹ the world's data will grow to 175 zettabytes in 2025. Let's dwell on that for a second: 175 zettabytes. What does that even mean? According to IDC's 'Data Age 2025' paper, if you were to store 175 zettabytes on DVDs, your stack of DVDs would be long enough to circle Earth 222 times. Almost every action we take leaves a digital trail – browsing online, shopping in a bricks-and-mortar store with a

credit card, sending an email, taking a photograph, reading an online article, even walking down the street if you're carrying a mobile phone or there are CCTV cameras in the vicinity.

If we learn how to read it properly, this huge amount of data can be put to use to do pretty much anything we want. This book is about how we go about applying data to drive business success. Data in itself isn't a new invention. Going back even before computers and databases, we still used data to track actions and simplify processes – think of paper transaction records and archive files. Computers, and particularly spreadsheets and databases, gave us a way to store and organize data on a large scale, in an easily accessible way. Suddenly, information was available at the click of a mouse.

The difference today is that the most valuable data we have available is different now. Unstructured data is information captured in its 'raw' form from the world around us. Pictures, videos, maps, text, speech recordings, social media posts – the list goes on. This data is potentially far more valuable than the business data we've worked with in the past because it can tell us a lot more about what we need to know.

Imagine you run a shop and you want to know more about the people walking past – your potential customers. You could quite easily survey the situation and create data – counting the number of people that walk past, and perhaps ticking a box if they are male or female, old or young. This would be structured data, and for a simple analysis it would be fine.

What if you wanted to know more about these people? How are they acting? Are they stopping to look in shop windows or walking quickly as if in a hurry to be somewhere else? Manually recording all of this data about every person on the street wouldn't be practical; it would take too long. So you can set up a video camera. Your data would be unstructured – you wouldn't be able to easily upload it to a spreadsheet and analyse it. But that value would be there, locked inside the data, if you had a way to get it out.

There's no doubt that the sheer amount of data we're creating is, well, big. But, if I'm honest, I've never been entirely comfortable with the term 'big data'. It feels too simplistic to me, focusing on the volume of data rather than the incredible opportunities this data creates. I wish there was a better term to describe this huge shift in our technology, culture and world. That's why, in this book, I talk about 'data' in all senses, big and small – because it doesn't matter how much data you have, it's whether you have the right data and are able to use it successfully that counts.

A brave new (data-driven) world

Thanks to your share of global data creation, quite a lot is known about you! It goes way beyond Google knowing what you've searched for online and Facebook knowing who you're friends with. Your Internet service provider knows every website you've ever visited. Ever. Even in private browsing. Google knows your age and gender (even if you've never told them) and you can be sure they have a comprehensive profile of you and your interests, so they can decide what ads to show you. Facebook clearly knows who you're friends with and who you're in a relationship with. But did you know Facebook can also predict whether your relationship is going to last or, if you're single, when you're about to be in a relationship (and with whom)? Facebook can also tell how intelligent you are, based on an analysis of your 'likes'. (In case you're wondering, liking curly fries, science, Mozart, thunderstorms or *The Daily Show* predicted high intelligence, while likes for Harley Davidson, Lady Antebellum and I Love Being a Mom predicted low intelligence.)

The police know where you're driving, certainly in the UK, where they have access to thousands of networked CCTV cameras across the country that scan number plates and take pictures of cars and their drivers. In the United States, many cities make similar use of traffic cameras. Your phone also knows how fast you're driving. For now, that information is kept well guarded and is not routinely shared with, for example, police, who might be able to use it to prosecute you if it shows you have broken laws. But more and more insurance companies are starting to make use of smartphone data to deduce who is a safe driver and who's a riskier prospect.

Your grocery store loyalty card tracks the brands you like and collects mountains of information on your purchasing habits and preferences. Retailers use this data to personalize your shopping experience, but it can also be used to predict what else you might want to buy in future.

In what has become one of the best-known and often-cited examples of businesses evolving their ability to analyse data, in 2012 US retailer Target showed how it could predict a teenage girl was pregnant (based on her buying habits) and start sending her baby-related offers – before the girl's family even realized she was pregnant. Today, things have moved on, and businesses like Amazon are moving towards being able to predict what we want to buy with strong enough certainty that they will ship orders towards us before we even place them.

Today, data analytics powers much more than ecommerce and targeted advertising, though. Its influence stretches to almost every aspect of modern life, from healthcare to space exploration, even to our politics.

Elections are increasingly driven by analytics, and since Barack Obama's victory in 2012, candidates have spent increasingly large sums of money on predicting how we will vote, so they can focus campaign resources where they will make a difference – on undecided and swing voters. Obama employed a team of more than 100 data analysts to run 66,000 computer simulations every day.

They collected and amalgamated all the data they could from voter registration data, donations, public records and bought-in third-party commercial data (including data mined from social media). Then everybody who had been identified was evaluated on their likelihood of voting for Obama, based on how well their data profile matched that of known supporters. Armed with their sophisticated demographic information, the team then launched targeted campaigns. These were aimed at increasing voter turnout and registration among sectors where the likelihood of backing their candidate was high, and influencing voter choice in sectors where the support metric indicated voters could go either way. This meant that targeted messages could be sent out by email, social media and browser display ads – depending on whether an individual needed to be convinced to register, vote or pick the correct candidate. In the years since then, all parties and most candidates have enthusiastically launched their own analytics strategies. Donald Trump's successful campaign, led by Jared Kushner, grew daily donations from \$8,000 per day to \$80,000 per day with targeted social media advertising. And the referendum that led to the UK's historic decision to leave the European Union was heavily analytics-driven. Campaigners on the 'leave' side developed sophisticated methods of profiling voters on Facebook (adopted from those developed by the Trump campaign) which were used to identify areas where support was strongest. To do this they used psychological mapping techniques developed by Michal Kosinski. Kosinski discovered that by analysing just 10 of a person's Facebook 'likes', he was able to correctly predict specific facts about that person, such as their age, income group and political affiliation, better than the average human could. By analysing 70 'likes', he was able to answer more questions about that person than a friend of theirs could.²

Data analytics is also helping to answer the question of whether there has ever been life on Mars. NASA's Jet Propulsion Laboratory, which runs the day-to-day mission planning for the Mars Rover spacecraft, is now using

Elasticsearch technology (also used by companies like Netflix and Goldman Sachs) to process all of the data transmitted from the Rover craft during its four daily scheduled uploads. While mission-planning decisions used to be based on the previous day's data, the move to real-time analytics vastly speeds up the time in which decisions can be taken by mission control. Patterns and anomalies in the datasets can be spotted far more quickly, and correlations which could provide mission-critical insights are more likely to become apparent, leading to a greater rate of scientific discovery and less danger of malfunction or failure.

As I write the second edition of this book, I can't ignore the fact that in the time that has passed since I wrote the first edition, the world has gone through unprecedented and previously unimaginable change. This has brought healthcare to the forefront of the data revolution, and undoubtedly a lot of the progress we've made in the fight against Covid-19 has been down to the vast increase in data we are generating, and our ability to analyse it.

For years, the basis of most medical research and discovery has been the collection and analysis of data: who gets sick, how they get sick and why. But now, with sensors in every smartphone and doctors able to share information across disciplines, the quantity and quality of the data available are greater than ever before, which means that the potential for breakthroughs and change is growing just as exponentially. Smartphones and other popular smart devices – including Jawbone, FitBit and others – now have the capacity to help people track their progress towards a healthier lifestyle. Today, many people wear watches on their wrist that can instantly give us an ECG test or analyse our levels of blood oxygenation. Computer vision (which we will explore later in the book) is used to scan medical images such as MRI and x-ray images, and spot symptoms of disease. It is at the point where it can do this as accurately as a newly trained human doctor now – but much, much faster, allowing it to review thousands of images in minutes. As well as Covid-19, data, AI and analytics are being used to track, analyse and treat epidemics around the world including Ebola and Zika.

All this is just the tip of the iceberg, and data volumes will only continue to grow. More often than not, when we sign up to a new product or service, whether it's a fitness tracker or a store loyalty card, we're happily giving access to our personal data – in return for benefits like improving our fitness or collecting points towards a free coffee. To me, signs indicate that as a society we are getting happier with the idea of handing over our data – as long as we are sure we are getting benefits from it (there are some important

caveats here, which will not be ignored as we go through this book). As more companies tap into the possibilities of data, and as the technology advances to gather more and more information, the amount of data available is predicted to grow exponentially.

We will also get better at analysing these heaps of data. The most powerful tools we have to do this today come under the heading of artificial intelligence (AI), and this book is strongly focused on harnessing this technology for business growth. Tech giants including Google, Amazon, Salesforce and IBM are all heavily invested in providing platforms that make this technology accessible to anyone. There's good reason for that, as worldwide sales of AI-enabling software, hardware and services are forecast to hit \$327 million in this year (2021) and increase to over \$550 million by 2024. That's a drop in the ocean compared with how much value will be generated by businesses putting those tools and services to work – which is expected to be in the trillions. But, during a gold rush, everyone knows it's always safest to be the one selling the shovels!

The other core technology paradigm shift driving this growth in data and analytics is the Internet of Things (IoT), sometimes known as the Internet of Everything (IoE). The IoT refers to devices that collect and transmit data via the Internet, and covers everything from your smartphone, smartwatch, FitBit band, even your TV and refrigerator. The IoT has seen enormous growth in recent years, and it's only just getting started. Today, there are about 21.5 billion devices that connect to the Internet. By 2024, that number is predicted to rise to over 80 billion. Smartphone users alone are predicted to number over 6 billion by 2020. By 2026, the value of the global market for this technology will reach \$1.6 trillion.

Smart devices are transforming our world, our cars, our homes and our businesses. What was once science fiction is already becoming reality – self-driving cars are clocking up thousands of miles a week and will be a regular sight on our roads soon, according to Tesla CEO Elon Musk.

'Wearable' technology is a crucial part of the IoT, and the global market for wearable devices has also ballooned. Twenty-one per cent of American consumers own or wear a smart watch or fitness tracker, as of 2020. All of these devices create a wealth of data, and we're only just starting to realize the implications of this now.

Connected devices can not only connect to the Internet; they can also connect and share information with each other. In fact, machine-to-machine connections will grow to 27 billion by 2024. So, in the near future, it's not unreasonable to imagine your refrigerator knowing when your milk is out of date and automatically telling your smartphone to order more in the next online shop.

Are we nearing true artificial intelligence?

In computing terms, artificial intelligence (AI) has been the ultimate goal since the very first computers were invented. It's also been a tantalizing prospect for science fiction writers! But are we finally getting close to realizing AI?

Combining cognitive science (the study of the human brain) and computer science, artificial intelligence is already impacting almost every area of our lives, from business to healthcare and even our private lives. The aim is to allow a computer to simulate human thought and mimic how our brains work. This allows computers to undertake things that we humans take for granted, like understanding natural language or recognizing objects in a picture.

The voice assistants that we see everywhere today are a prime example of current AI. The system 'learns' as it processes information, so the more data the system is given, the more it learns, and the more accurate it becomes. In practical terms, this technology could be used in any field in which a large amount of complex data needs to be processed and analysed to solve problems, including healthcare, law, education, finance and, of course, business. The technology is already being used in many industries and in our homes, through AI-enabled chatbots like Alexa and Siri.

As computers are more able to think like humans, they enhance our knowledge and capabilities. Just as the heroes of science fiction movies turn to their computers for analysis, predictions and conclusions on what to do next, in real life we're moving into an era where computers can enhance human knowledge in entirely new ways.

Today when we talk about AI, we often mean machine learning. This technology means computers can change and improve their algorithms by themselves, without being explicitly programmed by humans. We will move on to cover this terminology in more depth as we go through the book. For now, it's useful to remember that AI is a concept (intelligent machines), machine learning is a technology designed to provide AI, and other terms we will cover – supervised and unsupervised learning, reinforcement learning and deep learning – are categories of machine learning.

Is 'true' artificial intelligence just around the corner? Probably not – at least, not if by 'true' AI we mean the intelligent robots we're used to seeing in science fiction. Many scientists believe computers will never be able to 'think' like a human brain. But, however you want to frame it, computers' abilities to see, understand and interact with the world around them are growing at an incredible rate. And as the amount of data we have continues to increase, so too will computers' abilities to learn, understand and react.

The technology has advanced to such an extent that it is now possible for computers to recognize and respond to human emotions. Known as ‘affective computing’, this technology analyses facial expressions, posture, gesture, tone of voice, speech and even the rhythm and force of keystrokes to register changes in a user’s emotional state.

Imagine the potential of this technology. Your computer could recognize when you are frustrated or struggling with a task and provide additional information to help you along. Your phone could tell you to take a break when your stress levels are high. Or, without being asked, your smart home could provide soothing music and lighting when you get in from a bad day at the office. If this all sounds a bit far-fetched, it’s not. Leading organizations like Disney, the BBC and Coca-Cola have partnered with Affectiva, a company specializing in facial recognition technology, to test the effectiveness of adverts and assess how viewers react to content. The same company is also working with a Japanese car company to create in-car technology that can detect when you’re distracted or drowsy, and contact emergency services or your next of kin in the event of an emergency. Microsoft has even tested a bra that can sense stress levels in women.

Just as computers can never learn to ‘think’ in the same way as a human brain, these emotional machines will never really be emotional, but we’re nearing the time when machines will at least appear to give us suitable emotional responses. The really exciting part is that we’re only just starting to explore the possibilities of all this technology. In 10 years’ time, if not before, AI and machine learning will be thoroughly mainstream, and capable of things we can’t even imagine today!

The fourth industrial revolution – or Industry 4.0

Consensus opinion tells us that this explosion of data and analytics capabilities heralds a new technological era in the history of humankind. First came steam and the early machines that mechanized some of the work our ancestors did. Next came electricity, the assembly line and the birth of mass production. The third era of industry came about with the advent of computers and the beginnings of automation, when robots and machines began to replace human workers on those assembly lines.

Now what is being called the fourth industrial revolution – or Industry 4.0 – is well under way. This era brings together all of the advances of its predecessors with new developments of its own, giving rise to machines that operate as though they are capable of thinking and learning for themselves.

Are they actually thinking and learning? Well, those are terms that until now we've exclusively used in relation to organic, living beings, such as ourselves. But, in many ways, it's irrelevant. They are capable of carrying out tasks as if they are – and to put them to use in changing the world for the better, that's all we need!

Industry 4.0 brings together computers and automation in an entirely new way, with robotics connected remotely to computer systems equipped with machine learning algorithms that can learn and control the robotics with very little input from human operators.

It allows us to build 'smart factories', in which cyber-physical systems (a combination of computers, networks and physical actions) monitor the physical processes of the factory and make decentralized decisions. The machines are augmented with web connectivity and connected to a system that can visualize the entire production chain and make decisions on its own. They essentially become IoT systems, communicating and cooperating both with each other and with humans in real time via the wireless web.

All the data we have available, and the sophisticated systems that we can use to analyse it, has led to the emergence of the 'digital twin' process. This is a simulated version of just about any system, built using real-world data. For example, a digital twin of an Industry 4.0 factory will show all of the machinery, all of the working processes, and even information about the human workers on the factory floor. All of this data is captured by sensors and cameras in the real world and digitized, to let us simulate the results of changing any variable that is involved. This provides us with predictions that can be the basis of real-world decision-making. Will increasing the operating speeds of machinery lead to greater output, but increased component failure due to wear and tear? How will switching machinery or components for newer models impact overall costs and performance? What elements are likely to break down most frequently, therefore requiring maintenance and replacement parts to be on hand? Being able to accurately answer questions like these leads to businesses being less wasteful, more productive and ultimately more profitable.

As with any major shift in industry, there are challenges in adopting this approach. Data security issues are increased when you introduce new systems and increase accessibility. Systems need to be incredibly reliable and stable for successful cyber-physical operations, and this can be difficult to achieve and maintain, especially when you consider there is a systematic lack of experience and manpower to create and implement these systems (we cover this 'skills gap' in Chapter 14). Likewise, avoiding technical problems that

could cause expensive production outages is always a concern. In addition, there could be issues with maintaining the integrity and quality of the production process with less human oversight. Finally, whenever new automations are introduced, there is always a risk of losing valuable human jobs. All these issues combined with a general reluctance from stakeholders and investors to invest heavily in costly new technologies means Industry 4.0 has many hurdles to overcome before it becomes mainstream.

But the benefits of an Industry 4.0 model could outweigh the concerns for many production facilities. For example, in very dangerous working environments, the health and safety of human workers could be improved dramatically. Supply chains could be more readily controlled when there is data at every level of the manufacturing and delivery process. Computer control could produce much more reliable and consistent productivity and output. And the results for many businesses could be increased revenues, market share and profits.

In his book *The Fourth Industrial Revolution*,³ Professor Klaus Schwab, Founder and Executive Chairman of the World Economic Forum, describes how this fourth revolution is fundamentally different from the previous three, which were characterized mainly by advances in technology. In this fourth revolution, we are facing a range of new technologies that combine the physical, digital and biological worlds. These new technologies will impact all disciplines, economies and industries, and even challenge our ideas about what it means to be human. These technologies have great potential to continue to connect billions more people to the web, drastically improve the efficiency of businesses and organizations, and help regenerate the natural environment through better asset management, potentially even undoing all the damage previous industrial revolutions have caused.

The question, then, is not if Industry 4.0 is coming, but how quickly. I suspect that the early adopters will be rewarded for their courage jumping into this new technology, and those who avoid change risk becoming irrelevant.

Other world-changing technologies

As well as the vast increase in data, the arrival of AI and the rollout of the Internet of Things, several other new technologies are proving that they have a key role to play in the era of Industry 4.0.

One of these is the field of genomics – the field of applying technology to our understanding of the genome and processes such as gene editing, which are being used to create new medical treatments and possibly even eradicate some diseases outright. One method of gene editing known as CRISPR-Cas9 has been shown to have tremendous potential, and the technique is being used to reduce the risks posed by hereditary conditions as well as create crops that are more resistant to pests and fungus, and create allergen-free versions of food like nuts and gluten. One project is even focused on creating ‘super horses’ that can run faster and jump further than normal horses!

Another is blockchain – a form of database file secured by encryption, as well as the fact that it is stored on multiple locations that are all kept synchronized with each other and can’t be altered by anyone without permission. Blockchains can be used to create tamper-proof transaction ledgers and are used to track inventory, distribution and logistics and sales records. A World Economic Forum report predicted that by 2025, 10 per cent of GDP will be stored on blockchains⁴ – making it something that every business leader should at least be aware of.

Extended reality (XR) covers the concepts of both virtual reality (VR) and augmented reality (AR). By putting on a headset, or looking through a camera lens, we can step into entirely digital worlds (VR) or see elements of the digital world overlaid on our view of our own physical world. Already these technologies are widely used for education, entertainment, training and operational purposes.

None of these technologies exist or were developed in a vacuum, and in fact it’s true to say that none of them could have advanced to the stage they’re at now if other technologies – data technology, AI and IoT – weren’t already in place and developed to a certain level of sophistication themselves.

Why every business must become a data business

It’s clear that data is becoming a key business asset, central to the success of every company. As the world becomes smarter and smarter, data becomes the key to competitive advantage, meaning a company’s ability to compete will increasingly be driven by how well it can leverage data, apply analytics and implement new technologies.

Data and the ability to turn data into business value will become increasingly important in every sector within a few very short years. In fact, according to the International Institute for Analytics, businesses using data

saw \$430 billion in productivity benefits over competitors who are not using data between 2017 and 2020.⁵

In business, information is power, and data analytics is providing information we couldn't have dreamed of collecting or analysing just a few short years ago. Companies that don't evolve and embrace the data revolution have already been left behind. This is demonstrated starkly by the response from businesses to the Covid-19 pandemic. Those that built their business models around data – such as tech giants like Google and Amazon – have achieved record results at a time when the world is in turmoil, due to data giving them the resilience to adapt and evolve their processes and operations. Those that haven't been able to transition into becoming knowledge-based organizations have not all fared so well.

It all starts with a data strategy

Which all brings me quite nicely to why we are here. The aim of this book is to start looking at ways that any business, of any size, in any industry, can start putting data, and advanced data technologies like AI and the IoT, to work.

Businesses of all shapes have been enthusiastically jumping on board the AI bandwagon for a while now, and this is perfectly understandable. Both tech giants and start-ups have shown it's possible to generate huge growth and build truly world-changing products and services when they approach their challenges with a 'data-first' mindset.

Unfortunately, it doesn't always work out! In fact, according to one estimate, up to 85 per cent of projects that involve businesses working with data end up as failures – either not providing ROI or unable to fulfil the purpose they are intended for.⁶

I have worked with many companies to implement data projects, and there's one lesson I've had drummed into me time and time again: data is nothing without strategy.

Put simply, what this means is that it's very easy to get dazzled by the technology itself and the amazing things that can be done with it. When it comes to analytics and AI, lots of people start out by focusing on the 'what' and the 'how' questions, when really what they need to be thinking is 'why'.

Why do you need to change something about how your business operates? Why do you need to understand more about your processes? This is why strategy is so important in all areas of business, but particularly so when it comes to data strategy. Never mind (for now) the fact that AI can be

used to generate growth in this metric or predict failure rates of this particular component. You need to understand why it's important to your business that it can do this. What are you hoping to achieve by doing it?

A key idea I want to get across in this book is that there are six main areas where using data can be put to work to improve your business.

These are decision-making, understanding customers and markets, making better products, making better services, improving your operations and monetizing your data. Whichever you choose, though – one or all six – it's vital that you put data to work in a way that is in line with your overall business strategy.

Industry-dominating behemoths such as Google, Facebook and Amazon have all been pioneers – not simply by collecting vast quantities of data, but by finding innovative ways to put it to use that aligns with their strategies for growth. Everything they do has the overall aim of getting more users onto their platforms.

Over the course of the next 15 chapters of this book, I will go through the entire process, starting with identifying 'why' you need to use data and moving on to explore the vast range of options that are available for getting the job done, the methods of analytics that are proving to be the most valuable, the regulatory concerns, ethical considerations and practical implementations. Most importantly, we will develop a plan and process for ensuring that everything you do is geared towards your company's goals.

Notes

- 1 Reinsel, D, Gantz, J and Rydning, J (2018) The digitization of the world: from edge to core, IDC, resources.moredirect.com/white-papers/idc-report-the-digitization-of-the-world-from-edge-to-core (archived at <https://perma.cc/7BYJ-GZKH>)
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- 3 Schwab, K (2017) *The Fourth Industrial Revolution*, Portfolio Penguin, New York
- 4 World Economic Forum (2015) Deep shift: technology tipping points and societal impact, www3.weforum.org/docs/WEF_GAC15_Technological_Tipping_Points_report_2015.pdf (archived at <https://perma.cc/RTK2-CTC5>)